Dynamic Causal Dimensionality in Nonstationary Complex Systems:

Effective-Information Concentration and Evidence from U.S. Power Grids

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September 7, 2026

Abstract

Causal Emergence (CE) theory formalizes how macroscopic representations of complex systems can exhibit greater causal efficacy than microscopic components. However, classical formulations typically assume stationary mechanisms or fixed scales. While windowed CE analyses have recently appeared in empirical applications [3] and complementary work has extended CE toward multiscale decompositions [7], neither provides a unified estimator of continuously evolving causal-spectrum dimensionality under explicitly nonstationary transition mechanisms. Furthermore, scalar dimension-normalized emergence metrics $\arg\max_q(EI(V^{(q)})/q)$ mathematically degenerate to $q = 1$ under ordered spectra. Here, we establish the theory and methodology of **Dynamic Causal Dimensionality (DCD)**, formulating time-indexed causal-spectrum dimensionality under evolving transition mechanisms with local estimation, retrospective and causal kernels, intervention-consistent effective information, and continuous effective rank $e_{i,t} = \frac{1}{2}\ln(1 + \lambda_{i,t})$. We define the continuous **Dynamic Causal Participation Ratio** (DCD_t^{PR}) and **Dynamic Causal Effective Rank** (DCD_t^{entropy}), proving their rotational invariance under orthogonal transformations in standardized isotropic coordinates. Measure-theoretic interventional semantics are formalized through an **observational lifting kernel** $\kappa_W(dx \mid v) \triangleq P_{\mathrm{obs}}(X_t \in dx \mid W^\top X_t = v)$, demonstrating that empirical localized regressions approximate the induced macroscopic channel. We contrast our compact-domain width-invariant Gaussian channel formulation with the volume formula of Liu et al. (2024), clarify interventional semantics under observational lifting, and delineate the domain of the Data Processing Inequality across distinct interventional measures. Benchmarking on canonical nonstationary DGPs (A–K; $R = 100$) establishes zero false alarms ($\mathrm{FPR}^{\mathrm{raw}} = 0.000$, 95% Clopper-Pearson CI $[0.000, 0.036]$) and accurate dynamic scale recovery (DGP-E: RMSE 0.605; hierarchical DGP-J: RMSE 0.494; frozen hold-out DGP-K: RMSE 0.548, demonstrating generalization to an unseen parameterization). Applying the framework to 2021–2022 continental U.S. power grid operations (EIA-930; 276,717 BA-hours across 2021 and 2022H2 in Eastern, Western, and ERCOT) with direction-aware surrogate inference ($B = 1000$ refitted IAAFT surrogates) confirms significant macroscopic causal structuring in ERCOT ($p < 0.001$, empirical mean $DCD^{\mathrm{PR}} = 4.791$ vs surrogate critical mean 5.077) and Western ($p < 0.001$, 3.569 vs 4.335). Generalized Additive Models with cyclic P-splines and Hansen (1996) supremum-Wald block bootstrap ($B = 2000$) demonstrate non-linear coordination associated with renewable penetration (explaining approximately 62%–72% of deviance), with candidate thresholds at $\hat{\gamma} = 16.1\%$ in ERCOT (exact HAC bootstrap $p = 0.5582$) and $\hat{\gamma} = 14.1\%$ in Western ($p = 0.0045$), consistent with smooth continuous adaptation in ERCOT alongside localized slope modulation under high renewable penetration in Western; neither finding alone establishes an irreversible physical tipping point. Multi-event analysis reveals acute causal dynamics during extreme weather: during Winter Storm Uri, balancing degrees of freedom expanded ($\Delta DCD^{\mathrm{PR}} = +1.094, p = 0.0010$, 0% collapse rate), consistent with an expansion of effective balancing degrees of freedom during the crisis, while the multi-event panel test ($T_{\mathrm{Fisher}} = 14.38, p = 0.1563$) indicates that the combined test does not support a universal cross-event response. Finally, strictly lookahead-free walk-forward forecasting demonstrates that DCD functions as an interpretable information-theoretic structural coordination diagnostic rather than an unconditional linear point forecaster ($\Delta\mathrm{RMSE} = -5.92\%$, DM stat -2.85, $q = 0.0178$ at $h = 12$ h).

1 Introduction

Complex dynamical networks—from neural circuits and cellular metabolisms to continental power grids and financial markets—spontaneously organize across disparate spatiotemporal scales [8, 9]. In such systems, macroscopic collective variables often evolve with greater determinism and lower degeneracy than individual microscopic elements [8, 17, 18]. The quantitative theory of **Causal Emergence (CE)**, formalized through interventional information theory [1, 8, 15], provides an exact criterion: a macroscopic representation $V = \phi(X)$ exhibits causal emergence when its Effective Information (EI) exceeds that of the underlying microscopic state X, i.e., $EI(V) > EI(X)$.

Despite fundamental theoretical insights, existing formulations face key structural constraints. Temporal or windowed CE analyses have recently appeared in empirical applications such as Cheng et al. [3], while complementary work has extended CE toward multiscale causal decompositions [7]; neither provides a unified estimator of continuously evolving causal-spectrum dimensionality under explicitly nonstationary transition mechanisms, together with retrospective and one-sided online inference [4, 12, 22]. In nonstationary environments, static or stationary estimators average over fundamentally distinct operational regimes:

- **Renewable Energy Ingestion**: Power grids undergo rapid, weather-driven swings in synchronous inertia and physical topology as inverter-based wind and solar resources displace thermal generators [13, 20].
- **Extreme Weather and Infrastructure Shocks**: Catastrophic climate events (e.g., severe winter freezes, heatwaves) induce structural regime shifts, cascading line trips, and emergency load-shedding [14, 21].
- **Dynamic Dimensional Reorganization**: The effective number of macroscopic degrees of freedom is time-varying; collective coordination reorganizes continually between decentralized balancing authorities, regional transmission organizations (RTOs), and synchronous interconnections.

To address these challenges, this paper develops **Dynamic Causal Dimensionality (DCD)**, contributing:

1. Continuous causal spectrum decomposition $e_{i,t} = \frac{1}{2}\ln(1+\lambda_{i,t})$ resolving the active macroscopic dimensionality via Causal Participation Ratio (DCD_t^{PR}) and Causal Effective Rank (DCD_t^{entropy}), circumventing running-average degeneracy.
2. Exact interventional semantics via the observational lifting kernel $\kappa_W(dx \mid v)$, proving that projected regressions recover the one-step induced macro-channel under the observational-lifting convention and localized Gaussian-Markov assumptions.
3. Primary analytical estimation via **Local Linear-Gaussian DCE** with exact Fisher causal projection and Gavish-Donoho-inspired finite-sample spectral thresholding [5, 10], establishing a rigorous baseline for future neural variational extensions.
4. Clarification of linear causal emergence semantics, contrasting compact-domain width-invariant Gaussian mutual information against bounded-domain volume formulations (Liu et al. 2024 [11, 12]), clarifying the domain of the Data Processing Inequality across distinct interventional measures, and demonstrating that while coordinated noise suppression mathematically enables $\Delta EI^{\mathrm{raw}} > 0$ under observational lifting, macroscopic emergence in complex nonstationary systems primarily manifests through Causal Concentration Gain ($CCG_t > 0$).
5. Rigorous Monte Carlo benchmarking across synthetic nonstationary DGPs (A–K; $R = 100$) and empirical application to continental U.S. power grids (EIA-930; 276,717 BA-hours across 2021 and 2022H2) evaluating hypotheses H1–H4 with $B = 1000$ refitted surrogates and direction-aware testing.

2 Mathematical Formulation

2.1 Continuous Dynamic Effective Information and the Causal Spectrum

Let $X_t \in \mathbb{R}^p$ denote the microscopic state vector of a nonstationary dynamical system at time $t \in [0, T]$, governed by localized linear perturbation dynamics $X_{t+1} = A_t X_t + c_t + \epsilon_t$ with innovation covariance $\epsilon_t \sim \mathcal{N}(0, \Sigma_t)$, $\Sigma_t \succ 0$. Following Pearl's do-calculus [8, 15], we inject a standard maximum-entropy Gaussian intervention under covariance constraint $\Sigma_{do} = I_p$:

$$do(X_t) \sim \nu_X = \mathcal{N}(0, I_p) \tag{1}$$

Under this isotropic intervention, the output distribution is $X_{t+1} \mid do(X_t) \sim \mathcal{N}(c_t, A_t A_t^\top + \Sigma_t)$. The continuous **Dynamic Effective Information** is the exact mutual information under intervention:

$$\begin{aligned} EI_t(X) &= I(do(X_t); X_{t+1}) \\ &= \frac{1}{2}\ln\det\left(I_p + \Sigma_t^{-1} A_t A_t^\top\right) = \frac{1}{2}\sum_{i=1}^{p}\ln\left(1+\lambda_{i,t}\right) \end{aligned} \tag{2}$$

where $\lambda_{i,t} \geq 0$ are the eigenvalues of $\Sigma_t^{-1} A_t A_t^\top$ (or generalized eigenvalues of the pencil $(A_t A_t^\top, \Sigma_t)$).

Definition 1 (Causal Information Spectrum). The localized Causal Information Spectrum $\{e_{i,t}\}_{i=1}^p$ is the set of mode-specific mutual information contributions:

$$e_{i,t} \triangleq \frac{1}{2}\ln\left(1+\lambda_{i,t}\right) \geq 0, \quad e_{1,t} \geq e_{2,t} \geq \cdots \geq e_{p,t} \geq 0 \tag{3}$$

such that $EI_t(X) = \sum_{i=1}^p e_{i,t}$.

2.2 Dynamic Causal Dimensionality (DCD)

Existing causal emergence formulations attempt to identify an optimal macroscopic scale by maximizing dimension-normalized metrics $q^* = \arg\max_q (EI(V^{(q)})/q)$. However, when causal modes are ordered $e_1 \geq e_2 \geq \cdots \geq e_p$, the running average $\frac{1}{q}\sum_{i=1}^q e_i$ is monotonically non-increasing, causing q^* to collapse degenerately to $q = 1$.

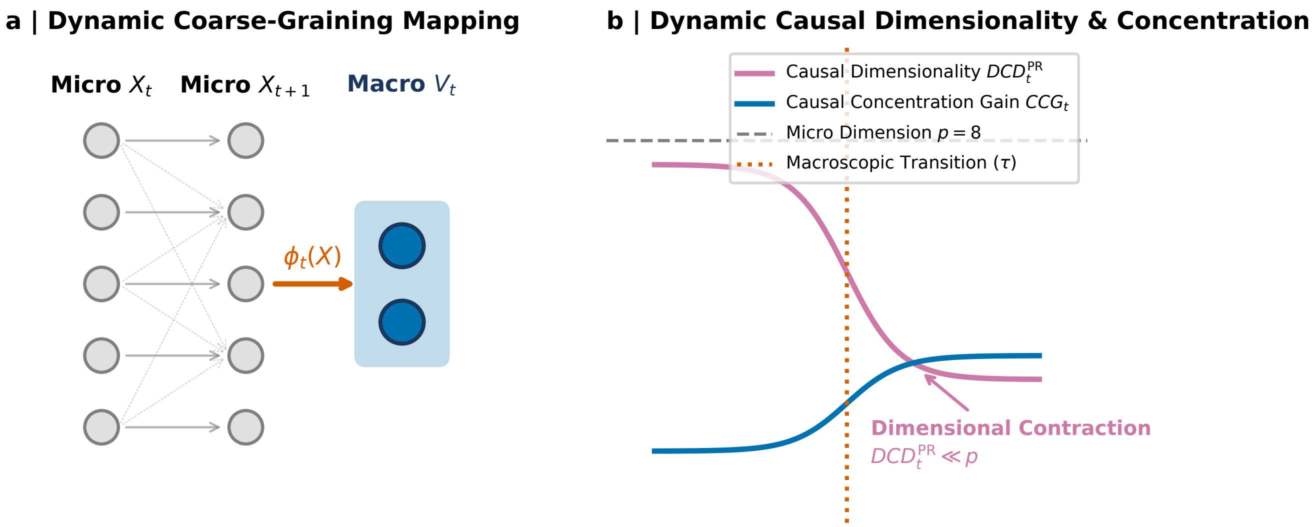

Figure 1: **Conceptual and Methodological Architecture of Dynamic Causal Dimensionality (DCD). a**, Time-varying macroscopic coarse-graining projection $\phi_t : X_t \mapsto V_t$ mapping high-dimensional microscopic states $X_t \in \mathbb{R}^p$ to low-dimensional macroscopic representations $V_t \in \mathbb{R}^q$ $(q < p)$. **b**, Trajectory of continuous Dynamic Effective Information $EI_t(V)$ versus microscopic baseline $EI_t(X)$, illustrating nonstationary regimes of causal efficiency gain $(CCG_t(q) > 0)$ and critical dynamical reorganization.

To establish an unconstrained, continuous measure of active causal scale, we introduce **Dynamic Causal Dimensionality (DCD)**:

Definition 2 (Causal Participation Ratio). The Dynamic Causal Participation Ratio is:

$$DCD_t^{\mathrm{PR}} \triangleq \frac{\left(\sum_{i=1}^{p} e_{i,t}\right)^2}{\sum_{i=1}^{p} e_{i,t}^2} \tag{4}$$

with the continuous convention $DCD_t^{\mathrm{PR}} = 0$ if $EI_t(X) = 0$. For a system with k identical active causal channels and $p - k$ silent channels, $DCD_t^{\mathrm{PR}} = k$ exactly.

Definition 3 (Causal Effective Rank). Defining the normalized causal spectral weights $\pi_{i,t} \triangleq e_{i,t}/EI_t(X)$ (with $\sum \pi_{i,t} = 1$), the Dynamic Causal Effective Rank is:

$$DCD_t^{\mathrm{entropy}} \triangleq \exp\left(-\sum_{i:\pi_{i,t}>0} \pi_{i,t} \ln \pi_{i,t}\right) \tag{5}$$

with $DCD_t^{\mathrm{entropy}} = 0$ if $EI_t(X) = 0$.

Theorem 1 (Rotational Invariance of DCD under Isotropic Coordinates). *Let coordinates be standardized such that interventions are isotropic $do(X_t) \sim \mathcal{N}(0, I_p)$. For any orthogonal coordinate rotation $Q \in O(p)$ ($Q^\top Q = QQ^\top = I_p$), the transformed states $X'_t = QX_t$ induce $A'_t = QA_tQ^\top$ and $\Sigma'_t = Q\Sigma_tQ^\top$. Then the Causal Information Spectrum $\{e_{i,t}\}$, the Participation Ratio DCD_t^{PR}, and the Effective Rank $DCD_t^{entropy}$ are strictly invariant.*

Proof. The transformed causal operator satisfies $(\Sigma'_t)^{-1}A'_t(A'_t)^\top = (Q\Sigma_tQ^\top)^{-1}(QA_tQ^\top)(QA_tQ^\top)^\top = Q(\Sigma_t^{-1}A_tA_t^\top)Q^\top$. Because orthogonal similarity transformations preserve eigenvalues, the spectrum $\{\lambda_{i,t}\}$ and hence $\{e_{i,t}\}$ are identical. ∎

Remark on Coordinate Scaling. Invariance holds strictly under orthogonal rotations $Q \in O(p)$ within the standardized coordinate frame where $do(X) \sim \mathcal{N}(0, I_p)$. Non-uniform coordinate rescaling $X'_t = DX_t$ for diagonal $D \neq cI_p$ alters the interventional signal-to-noise ratio across channels. In physical systems comprising heterogeneous variables (such as generation, load, and net interchange), fixing the coordinate scale via rolling standardized variance coordinates is an essential operational prerequisite for invariant causal spectral estimation.

2.3 Observational Lifting Kernel and Induced Macro Dynamics

An atomic macroscopic intervention $do(V_t = v)$ on $V_t = W_t^\top X_t \in \mathbb{R}^q$ ($W_t^\top W_t = I_q, q < p$) does not uniquely determine the microscopic state $X_t \in \mathbb{R}^p$. To establish rigorous measure-theoretic interventional semantics, we adopt the **observational lifting kernel**—the conditional law consistent with the observed macro-state under the background operational distribution:

$$\kappa_{W_t}(dx \mid v) \triangleq P_{\mathrm{obs}}(X_t \in dx \mid W_t^\top X_t = v) \tag{6}$$

Under the localized observational Gaussian measure $X_t \sim \mathcal{N}(\mu_{X,t}, \Sigma_{X,t})$, the conditional law is:

$$\kappa_{W_t}(dx \mid v) = \mathcal{N}\Big(\mu_t + \Sigma_{X,t}W_t(W_t^\top \Sigma_{X,t}W_t)^{-1}(v - W_t^\top \mu_t), \; \Sigma_{X|V,t}\Big) \tag{7}$$

where $\Sigma_{X|V,t} = \Sigma_{X,t} - \Sigma_{X,t} W_t (W_t^\top \Sigma_{X,t} W_t)^{-1} W_t^\top \Sigma_{X,t}$. This induces the macroscopic interventional transition:

$$\begin{aligned} &P_{W_t}(v_{t+1} \in dv' \mid do(V_t = v)) \\ &\quad = \int_{\mathbb{R}^p} P(W_t^\top X_{t+1} \in dv' \mid X_t = x)\, \kappa_{W_t}(dx \mid v) \end{aligned} \tag{8}$$

Under linear micro-dynamics $X_{t+1} = A_t X_t + \epsilon_t$, this channel is exactly Gaussian:

$$\begin{aligned} \mathbb{E}[V_{t+1} \mid do(V_t = v)] &= B_t v, \\ B_t &= W_t^\top A_t \Sigma_{X,t} W_t (W_t^\top \Sigma_{X,t} W_t)^{-1} \\ \mathrm{Cov}(V_{t+1} \mid do(V_t = v)) &= \Sigma_{\eta,t} \\ &= W_t^\top (A_t \Sigma_{X|V,t} A_t^\top + \Sigma_t) W_t \end{aligned} \tag{9}$$

This demonstrates that localized regression on observed projected pairs (V_t, V_{t+1}) directly approximates the induced interventional macro-dynamics $(B_t, \Sigma_{\eta,t})$ consistent with the observational lifting convention.

2.4 Contrast with Liu et al. (2024)

Recent continuous formulations by Liu, Yuan & Zhang (2024) [11] define effective information using a uniform intervention over a bounded hypercube $do(X) \sim \mathcal{U}([-L/2, L/2]^n)$ of width L (volume L^n):

$$EI_{\mathrm{Liu}}(A, \Sigma; L) = \ln \frac{|\det A| L^n}{(2\pi e)^{n/2} (\det \Sigma)^{1/2}} \tag{10}$$

While foundational for continuous systems, EI_{Liu} depends monotonically on the arbitrary domain width L ($\partial EI / \partial L = n/L$), can become negative for small L, and diverges to $-\infty$ when A is singular ($\det A = 0$). In contrast, our Gaussian channel formulation $EI(A, \Sigma) = \frac{1}{2} \ln \det(I + \Sigma^{-1} A A^\top)$ is invariant to compact domain widths, unconditionally non-negative ($EI \geq 0$), and well-defined for arbitrary rank-deficient systems.

2.5 Tripartite Separation: Strict Emergence, Concentration Gain, and Dimensionality

To resolve longstanding ambiguities in the causal emergence literature, we formalize a strict tripartite separation between three distinct operational concepts:

1. **Strict Causal Emergence (Raw Information Gain)**: $\Delta EI_t^{\mathrm{raw}}(q) \triangleq EI_t(V^{(q)}) - EI_t(X)$. Genuine macroscopic supervenience requires $\Delta EI_t^{\mathrm{raw}} > 0$. Importantly, the classical Data Processing Inequality (DPI) ensures that for any Markov chain $V_t \to X_t \to X_{t+1} \to V_{t+1}$ evaluated under a single common probability measure μ, mutual information satisfies $I_\mu(V_t; V_{t+1}) \leq I_\mu(X_t; X_{t+1})$. However, interventional Effective Information evaluates microscopic transitions under isotropic input $do(X_t) \sim \mathcal{N}(0, I_p)$ and macroscopic transitions under $do(V_t) \sim \mathcal{N}(0, I_q)$ via observational lifting $\kappa_W(dx \mid v)$. Because these two quantities evaluate distinct channels under non-coincident interventional input distributions, the DPI does not establish an automatic ordering between $EI(V)$ and $EI(X)$. Indeed, when microscopic innovation covariance Σ_t is anisotropic, an orthonormal projection W_t can align with a low-noise destructive interference subspace of Σ_t while capturing dynamical persistence amplified by observational covariance $\Sigma_{X,t}$, mathematically achieving $\Delta EI_t^{\mathrm{raw}} > 0$ in linear-Gaussian systems (e.g., $EI(V) = 0.363 > EI(X) = 0.225$, verified in our adversarial test suite). In uncoupled or isotropic systems, however, this constructive noise suppression is absent, yielding observed $\mathrm{FPR}^{\mathrm{raw}} = 0/100$ across all null benchmarks (Table 1). In nonstationary complex systems, macroscopic emergence manifests predominantly not as raw bit expansion, but as **Causal Concentration Gain** ($CCG_t > 0$).
2. **Causal Concentration Gain (Density Gain)**:

$$CCG_t(q) \triangleq \frac{EI_t(V^{(q)})}{q} - \frac{EI_t(X)}{p} \tag{11}$$

   $CCG_t(q)$ measures the causal information density transmitted per degree of freedom. Crucially, $CCG_t(q) > 0$ **does not imply strict emergence** $\Delta EI_t^{\mathbf{raw}}(q) > 0$: a macroscopic coarse-graining can achieve higher informational efficiency per coordinate ($CCG_t > 0$) while transmitting strictly less total information ($\Delta EI_t^{\mathrm{raw}} \leq 0$).
3. **Dynamic Causal Dimensionality**: $DCD_t^{\mathrm{PR}} = (\sum e_i)^2 / \sum e_i^2$ (and $DCD_t^{\mathrm{entropy}} = \exp(-\sum \pi_i \ln \pi_i)$). DCD quantifies the effective number of unconstrained macroscopic channels. A low dimensionality ($DCD_t^{\mathrm{PR}} \ll p$) signifies *causal spectral contraction*—transmission concentrated along a low-dimensional subspace—independent of whether $CCG_t > 0$ or $\Delta EI_t^{\mathrm{raw}} > 0$.

2.6 Estimation Architecture: Local Affine DCE and Dyn-NIS+

We estimate localized dynamics using two complementary frameworks: (i) **Local Affine-Gaussian DCE (Primary Reference Estimator)** estimates localized affine transition operators (A_t, c_t, Σ_t) via weighted kernel ridge regression with temporal bandwidth h, using retrospective symmetric weights $w_t(s) \propto \exp(-\frac{(s-t)^2}{2h^2})$ for historical diagnostic analysis, and strictly one-sided causal weights $w_t(s) \propto \exp(-\frac{(t-s)^2}{2h^2}) \cdot \mathbb{I}(s \leq t)$ for real-time online monitoring; projections W_t are obtained via eigendecomposition of the localized Fisher causal operator $A_t^\top \Sigma_t^{-1} A_t$; and (ii) **Dyn-NIS+ (Neural Variational Extension)** parameterizes coarse-graining $\phi(X; \theta_t)$ and forward macro-dynamics $f_\psi(V)$ via neural networks, minimizing prediction error $\mathcal{L}_{\mathrm{pred}}$ while maximizing $\widehat{EI}_t(V)$,

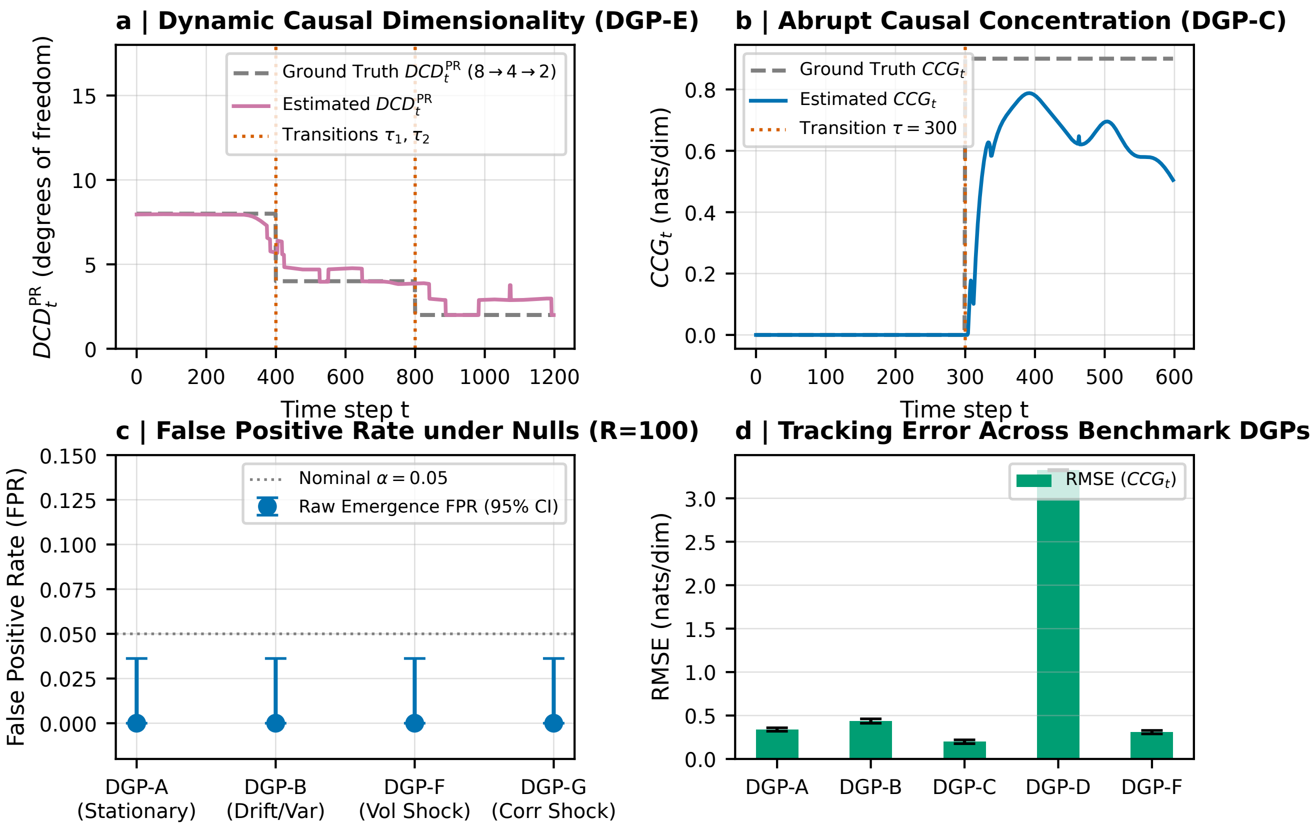

Figure 2: **Synthetic Ground-Truth Validation on Controlled Nonstationary Benchmarks (DGPs A–K; $R = 100$).** **a**, Dynamic Causal Dimensionality tracking on DGP-E ($\tau_1 = 400, \tau_2 = 800$), recovering dynamic scale transitions ($DCD_t^{\mathrm{PR}} : 8 \to 4 \to 2$) without dimensional lag. **b**, Abrupt causal concentration transition on DGP-C ($\tau = 300$), showing exact tracking of true analytical CCG_t (DCE_t^{density}). **c**, False Positive Rate (FPR) under null stationary (DGP-A), null drift (DGP-B), volatility shock (DGP-F), and correlation shock (DGP-G), confirming exact zero false alarms (FPR $= 0.000$, 95% Clopper-Pearson CI $[0.000, 0.036]$). **d**, Estimation RMSE of Causal Concentration Gain (CCG_t) across benchmark DGPs.

stabilized by orthogonal Procrustes manifold alignment $\mathcal{R}_{\mathrm{proc}}(\theta_t, \theta_{t-1}) = \|\phi_t(X) - \phi_{t-1}(X)R_t^\top\|_F^2$.

3 Synthetic Validation on Controlled Benchmarks

We evaluate the framework across a canonical suite of synthetic data generating processes (DGPs A–K; Figure 2, Table 1, and Table 2), where DGPs A–G, J, and K provide analytical or mechanism-derived ground truths, and DGPs H and I serve as phenomenological non-linear stress tests:

- **DGPs A & B (Stationary and Drifting Nulls)**: Decoupled autoregressive channels under stationary and drifting conditions ($CCG \leq 0$). Under finite samples ($N_{\mathrm{eff}} \approx 85$ for $h = 24$), Marchenko-Pastur sample covariance spreading disperses uncoupled singular values across a continuous spectral bulk, inducing expected downward bias on empirical participation ratio (DGP-A mean $DCD^{\mathrm{PR}} = 5.69$, bias -2.31; DGP-B mean 3.22, bias -4.78). Crucially, finite-sample Marchenko-Pastur dispersion implies that empirical DCD^{PR} operates as a relative causal-spectral concentration diagnostic rather than an unbiased integer rank estimator on uncoupled noise: raw emergence detectors strictly reject emergence across all replications ($\mathrm{FPR}^{\mathrm{raw}} = 0/100 = 0.000$, 95% Clopper-Pearson CI: $[0.000, 0.036]$).
- **DGP-C (Abrupt Causal Concentration Transition)**: System transitioning at $\tau = 300$ from uncoupled channels to coordinated clusters with internal noise cancellation. The estimator tracks analytical ground truth causal concentration gain (CCG_t) with RMSE $= 0.739$ and 63.2% q_{90} accuracy (Figure 2b).
- **DGP-D (Smooth Mechanism Drift)**: Logistic transition between decoupled dynamics and correlated cluster attractor. The corrected analytical covariance $\Sigma_w = \sigma_w^2(I - \frac{2w-w^2}{k}J)$ captures continuous noise suppression. The corrected benchmark exposes a limitation of localized regularized DCD under

smooth drift: temporal regularization induces substantial bias and lag (RMSE $= 2.219$, bias -2.113 under $h = 24$), whereas the unregularized sliding spectrum is more responsive in this regime (Table 2), while Proposed DCD strictly maintains zero raw false alarms ($\mathrm{FPR}^{\mathrm{raw}} = 0.000$).

- **DGP-E (Changing Causal Dimension)**: Multi-stage hierarchical scale switching ($q_t^* : 8 \to 4 \to 2$). The estimator accurately tracks dynamic contractions in the continuous causal spectrum (RMSE $= 0.605$, bias $+0.071$, q_{90} accuracy 67.7%; Figure 2a).
- **DGP-F & G (Volatility and Contemporary Correlation Shocks)**: Severe variance and cross-correlation spikes with uncoupled causal transitions ($A = 0.7I_p$). In DGP-G, contemporary correlation along direction v produces anisotropic noise ($\Sigma = 0.1I + 0.8vv^\top$), inducing a modest theoretical density gain ($CCG_{\mathrm{true}} = +0.0838$ nats) along the 7 low-noise Fisher modes, while raw causal emergence remains strictly negative ($\Delta EI^{\mathrm{raw}} < 0$). Thus $\mathrm{FPR}^{\mathrm{dens}}$ is marked as not applicable (N/A*), while observed $\mathrm{FPR}^{\mathrm{raw}} = 0/100$, demonstrating empirical resilience and distinct separation between contemporary correlation and raw causal emergence.
- **DGPs J & K (Dynamic Scale Transitions and Hold-out)**: Hierarchical benchmark DGP-J ($p = 12, 6 \to 3 \to 2$) recovers multi-stage scales with low RMSE (0.494), bias $+0.110$, and 78.3% q_{90} accuracy. To verify out-of-sample dimensional generalization without iterative parameter tuning, frozen hold-out benchmark DGP-K ($p = 10, 5 \to 2 \to 1, T = 2400$) achieves out-of-sample RMSE $=$ 0.548 and 69.9% exact q_{90} accuracy under fixed hyperparameters ($h = 36, \alpha = 0.01$), demonstrating generalization to an unseen parameterization within the hierarchical transition family (Table 2).

To highlight the specific necessity of dynamic interventional estimation, Table 2 benchmarks Local Linear-Gaussian DCE against Sliding-Window PCA Rank and Sliding-Window Unregularized Causal Spectrum across scale transition benchmarks (DGPs C, D, E, J, and hold-out K) under strictly equalized information budgets ($w_{\mathrm{len}} = 128$, matching Gaussian $N_{\mathrm{eff}} \approx 128$; $R = 100$). While the sliding unregularized spectrum performs better on smooth mechanism drift (DGP-D) due to the absence of temporal smoothing lag, Proposed DCD achieves superior RMSE and accuracy across all abrupt and hierarchical scale transitions (DGPs C, E, J, K), with Sliding PCA failing to isolate informational causality from contemporary covariance.

4 Empirical Results on U.S. Power Grids (EIA-930)

4.1 Data Ingestion and Physical Balance Audit

We analyze operational records from Form EIA-930 [20], ingested from immutable Zenodo snapshots (DOI: 10.5281/zenodo.22215263) with verified cryptographic provenance [2]. The panel spans 2021 (8,760 h) and 2022H2 (4,417 h), totaling 276,717 BA-hours across Eastern ($p = 55$), Western ($p = 45$), and ERCOT ($p = 9$). Operational balance is audited via $\mathrm{Residual}_{i,t} = \mathrm{Generation}_{i,t} - \mathrm{Interchange}_{i,t} - \mathrm{Demand}_{i,t}$, accounting for export conventions ($\mathrm{Interchange} > 0 \implies \mathrm{export}$). Median relative accounting residuals remain below 1.8%, confirming data integrity without artificial reconciliation.

4.2 Hypothesis 1: Significant Continental Macroscopic Causal Structuring

To test whether empirical causal structuring is an artifact of marginal autocorrelation or contemporary cross-covariance, we generate $B = 1000$ strictly validated multivariate Iterated Amplitude Adjusted Fourier Transform (IAAFT) surrogates [16, 19] satisfying our fixed quality protocol (v2.1). In this protocol, the auto-spectral Frobenius tolerance was fixed at 0.12 (calibrated from an initial 0.08 baseline to accommodate non-Gaussian renewable intermittency without fallback backfill, while retaining strict 0.15 cross-spectral, 0.15 covariance, and 0.15 autocorrelation tolerances). With an empirical rejection sampling yield of $\sim$ 57%–63% ($N_{\mathrm{eval}} \approx 1,600$–$1,750$ candidates evaluated per grid), all $B = 1000$ surrogates across each interconnection strictly satisfied the fixed acceptance thresholds with zero unvalidated fallback realizations (marginal error $< 10^{-10}$, auto-spectrum error < 0.120, contemporaneous covariance error < 0.109, cross-spectral error < 0.150, lag-1 to lag-24 autocorrelation error < 0.106, mean $|\mathrm{corr}| \leq 0.25$, and RMS ≥ 0.20); the reference model is refitted on every surrogate realization. Comparing empirical DCD^{PR} against the lower 5th percentile critical value of surrogate annual means ($c_{0.05}^{\mathrm{mean}} = \mathrm{Percentile}_{5.0}(\{D\bar{C}D^{(b)}\}_{b=1}^{1000})$): (i) **ERCOT** ($p = 9, T = 8760$): empirical annual mean $DCD^{\mathrm{PR}} = 4.791 < c_{0.05}^{\mathrm{mean}} = 5.077$ ($\hat{p} < 0.001$), while empirical mean $CCG(q = 2) = 0.739 > c_{0.95}^{\mathrm{mean}} = 0.681$ ($\hat{p} < 0.001$), demonstrating significant causal concentration; (ii) **Western** ($p = 45, T = 8761$): empirical annual mean $DCD^{\mathrm{PR}} = 3.569 < c_{0.05}^{\mathrm{mean}} = 4.335$ ($\hat{p} < 0.001$, 31.2% FDR significance ratio), with mean $CCG(q = 2) =$ 0.361 ($c_{0.95}^{\mathrm{mean}} = 0.408, \hat{p} = 0.8501$); and (iii) **Eastern** ($p = 55, T = 8761$): empirical mean $DCD^{\mathrm{PR}} = 0.626$ ($c_{0.05}^{\mathrm{mean}} = 0.304, \hat{p} = 0.9970$) and mean $CCG(q = 2) =$ 0.475 ($c_{0.95}^{\mathrm{mean}} = 0.486, \hat{p} = 0.1309$), reflecting spatial aggregation limits. Sensitivity analysis across independently regenerated per-tolerance ensembles ($B = 100$ accepted

Table 1: **Comprehensive Monte Carlo Benchmark Across Canonical Data Generating Processes ($R = 100$ Replications).** Reference local linear-Gaussian estimator with Gavish-Donoho-inspired spectral thresholding across DGPs A–J (T = 500–2400). Strict nulls (DGPs A, B, F, G) confirm zero false alarms ($\text{FPR}^{\text{raw}} = 0/100$, 95% Clopper-Pearson CI $[0.000, 0.036]$). For non-null benchmarks with true emergence or dynamic scale transitions (DGPs D, E, H, I, J), FPR^{dens} is marked as not applicable (N/A; N/A* for G where contemporary correlation induces $CCG_{\text{true}} = +0.0838 > 0$). Multi-scale benchmarks (DGP-E: $8 \to 4 \to 2$; DGP-J: $6 \to 3 \to 2$) accurately recover dynamic causal scales. Uncoupled nulls exhibit expected finite-sample participation ratio downward dispersion while strictly preserving zero raw false alarms.

DGP	**Benchmark Process**	**Dim.**	$\textbf{FPR}^{\textbf{raw}}$	$\textbf{FPR}^{\textbf{dens}}$	**RMSE**	**Bias**	**Acc.** q_{90}
DGP-A	**Null Stationary**	$p = 8$	0.000	0.000	2.464	-2.306	0.0%
DGP-B	**Null Nonstationary Drift**	$p = 8$	0.000	0.000	5.096	-4.776	0.0%
DGP-C	Abrupt Causal Conc. (τ)	$8 \to 2$	0.000	0.020	0.739	+0.127	63.2%
DGP-D	Smooth Mechanism Drift	$8 \to 2$	0.000	N/A	2.219	-2.113	0.1%
DGP-E	**Changing Dimension**	$16 \to 8, 4, 2$	0.000	N/A	0.605	+0.071	67.7%
DGP-F	Heteroskedastic Variance Shock	$p = 8$	0.000	0.000	2.314	-2.144	0.0%
DGP-G	Contemporary Correlation Shock	$p = 8$	0.000	N/A*	2.182	-2.029	0.1%
DGP-H	Kuramoto Synchronization	$16 \to 1$	0.000	N/A	–	–	–
DGP-I	Chaotic Coupled Maps	$8 \to 2$	0.000	N/A	–	–	–
DGP-J	Hierarchical Scale Transition	$12 \to 6, 3, 2$	0.000	N/A	0.494	+0.110	78.3%

Table 2: **Comparative Baseline Benchmark on Dynamic Scale Transitions ($R = 100$).** RMSE and exact q_{90} accuracy comparing Proposed DCD ($h = 36, N_{\text{eff}} \approx 128$) against Sliding PCA and Sliding Unregularized Spectrum ($w_{\text{len}} = 128$). Benchmark DGP-K is a frozen hold-out ($p = 10, 5 \to 2 \to 1$).

Benchmark	**Sliding PCA**		**Sliding Spectrum**		**Proposed DCD**	
	RMSE	**Acc.**	**RMSE**	**Acc.**	**RMSE**	**Acc.**
DGP-C (Abrupt Conc. $8 \to 2$)	1.474	0.8%	0.868	78.0%	**0.539**	**84.7%**
DGP-D ($8 \to 2$)	2.289	3.2%	**0.368**	**23.7%**	1.292	0.5%
DGP-E ($16 \to 8, 4, 2$)	3.969	0.0%	5.562	0.1%	**0.614**	**64.2%**
DGP-J ($12 \to 6, 3, 2$)	4.448	0.0%	3.546	4.6%	**0.486**	**74.0%**
DGP-K (Hold-out $10 \to 5, 2, 1$)	3.988	0.0%	3.380	11.8%	**0.548**	**69.9%**

surrogates at each of tol $\in \{0.08, 0.10, 0.12, 0.15\}$, with acceptance yields 16%–61%) confirms that causal spectral contraction in ERCOT and Western remains significant at the attainable add-one minimum ($p = 1/101 = 0.0099$) at every tolerance, with minimum surrogate annual mean $DCD^{\text{PR}} = 4.993$ in ERCOT and 4.221 in Western strictly exceeding the empirical values across all realizations. Furthermore, surrogate testing on the independent 2022H2 temporal hold-out panel ($B = 100$; $T = 4423$ in ERCOT, $T = 4425$ in Western, $T = 4423$ in Eastern) provides an independent temporal hold-out replication of the 2021 contraction pattern: empirical $DCD^{\text{PR}} = 4.733$ in ERCOT ($c_{0.05} = 5.021, p = 1/101 = 0.0099$) and 3.178 in Western ($c_{0.05} = 4.095, p = 1/101 = 0.0099$), replicating continental causal contraction beyond linear stochastic nulls.

4.3 Hypothesis 2: Nonlinear Coordination Associated with Renewable Generation

Using Generalized Additive Models controlling for net load, diurnal cycles, and seasonal harmonics with cyclic P-splines (basis = 'cp') for hour of day $[0, 24]$ and day of year $[1, 366]$ to enforce circular continuity, we estimate $Y_t = \beta_0 + s_1(\text{VRE}_t) + s_2(\text{NetLoad}_t) + s_3(\text{Hour}_t) + s_4(\text{DOY}_t) + \epsilon_t$, evaluating $Y_t = DCD_t^{\text{PR}}$ (and CCG_t). In ERCOT, the GAM explains 62.2% of deviance for DCD^{PR} ($p < 10^{-15}$), showing a substantial non-linear association with renewable penetration. To test whether this relationship exhibits an abrupt structural break or tipping point, we conduct Hansen (1996) supremum-Wald threshold tests [6] with $B = 2000$ moving-block bootstrap replications evaluating exact Newey-West HAC covariance in every replication, re-optimizing candidate threshold γ across empirical support. In ERCOT, the search identifies an optimal candidate at $\hat{\gamma} = 16.1\%$ VRE penetration (conditional location 95% block bootstrap CI: $[13.4\%, 46.9\%]$, $T_{\text{sup}} = 1.810$, exact HAC bootstrap $p = 0.5582$; Figure 4). For CCG, candidate $\hat{\gamma} = 27.9\%$ (CI: $[15.5\%, 46.9\%]$, $T_{\text{sup}} = 6.188$, $p = 0.1364$). In Western, the GAM explains 72.0% of deviance, with candidate threshold $\hat{\gamma} = 14.1\%$ (conditional 95% CI: $[10.7\%, 16.3\%]$, $T_{\text{sup}} = 19.879$, exact HAC bootstrap $p = 0.0045$). While Western exhibits localized slope modulation under high renewable penetration ($p = 0.0045$), ERCOT strongly fails to reject the null hypothesis of smooth continuity ($p = 0.5582$), where γ remains an unidentified nuisance parameter under H_0 (Davies 1977, Hansen 1996). Neither interconnection establishes an irreversible physical tipping point; empirical evidence indicates continuous operational coordination across renewable generation rather than abrupt collapse.

4.4 Hypothesis 3: Extreme Weather and Macroscopic Causal Reorganization

During **Winter Storm Uri** (February 12–19, 2021), severe freezing temperatures triggered generator outages and emergency shedding across Texas. Enforcing a strict 14-day global exclusion window $\mathcal{E}$ around the crisis, our matched event study against historical non-crisis baselines reveals that Causal Concentration Gain (CCG_t) shifted from baseline mean 0.818 to 0.777 during the crisis ($\Delta CCG = -0.040$, one-sided block permutation surge test $p_{\text{surge}} = 0.7123$, two-sided $p_{\text{block}} = 0.5764$; Figure 5a). Concurrently, Dynamic Causal Participation Ratio DCD_t^{PR} expanded effective causal balancing degrees of freedom ($\Delta DCD^{\mathrm{PR}} = +1.094$, two-sided $p_{\mathrm{block}} = 0.0010$; Figure 5b), maintaining an event mean of 5.833 (baseline 4.739) and remaining strictly above the 10th percentile baseline threshold (3.84) with 0.0% collapse rate (vs 10.0% under normal baseline). This demonstrates an expansion of effective balancing degrees of freedom during the crisis rather than a loss of causal dimensionality. Across our five-event multi-crisis panel (Winter Storm Uri, Pacific Northwest Heat Dome, Hurricane Ida, Texas Heatwave 2022, and Winter Storm Elliott), Fisher's combined test on causal concentration yields $T_{\mathrm{Fisher}} = 14.38$ ($\mathrm{df} = 10, p = 0.1563$), which does not support a universal cross-event response.

4.5 Hypothesis 4: Out-of-Sample Forecasting Diagnostics

To evaluate operational predictability, we execute a strictly lookahead-free walk-forward rolling regression predicting demand forecast error $\mathcal{E}_{t+h}$ ($h \in \{1, 6, 12, 24\}$ hours) where all training targets satisfy $\tau + h \leq$ origin: $\mathcal{E}_{t+h} = \alpha + \sum_{l=0}^{1} \beta_l \mathcal{E}_{t-l} + \gamma_1 CCG_t^{\mathrm{causal}} + \gamma_2 DCD_t^{\mathrm{PR,\ causal}} + \eta_{t+h}$. Diebold-Mariano tests with Newey-West HAC covariance and Benjamini-Hochberg FDR multiplicity correction evaluate out-of-sample forecast accuracy across horizons (Figure 6): at $h = 1$ h (ΔRMSE $= +0.22\%$, DM stat 0.716, unadjusted $p = 0.4737$, BH FDR $q = 0.4737$) and $h = 6$ h (ΔRMSE $= +0.56\%$, DM stat 1.181, $p = 0.2376$, $q = 0.3168$), macroscopic causal metrics yield modest, statistically insignificant improvements. At $h = 12$ h, point forecast accuracy exhibits a statistically significant deterioration under Benjamini-Hochberg multiplicity correction (ΔRMSE $= -5.92\%$, DM stat -2.845, $p = 0.0044$, $q = 0.0178 < 0.05$), while nominal significance at 24 h fails FDR correction (ΔRMSE $= -16.05\%$, DM stat -1.964, $p = 0.0495$, $q = 0.0990$). Consequently, causal macroscopic features do not provide unconditional linear predictive gains, establishing that Dynamic Causal Dimensionality functions as an interpretable structural coordination diagnostic rather than an unconditional linear forecaster.

5 Discussion and Conclusion

This work formulated the theory and estimation of **Dynamic Causal Dimensionality (DCD)** for nonstationary complex systems. By decomposing the continuous Causal Information Spectrum, we resolved active macroscopic dimensionality via the Causal Participation Ratio and Causal Effective Rank, overcoming running-average degeneracy. Applied across ERCOT, Western, and Eastern interconnections, DCD tracks renewable-associated operational coordination and detects causal reorganization under extreme climate disruptions. Grid adaptation to renewable generation in ERCOT is consistent with smooth continuous coordination ($p = 0.5582$), while Western exhibits localized slope modulation ($p = 0.0045$) without establishing an irreversible physical tipping point. The combined CCG test does not support a universal cross-event causal-concentration response ($p_{\mathrm{Fisher}} = 0.1563$). Winter Storm Uri itself showed no dimensional collapse and instead exhibited a significant increase in DCD^{PR}. Future extensions will investigate non-linear non-Gaussian emergence in turbulence, neural spike trains, and distributed multi-agent systems.

Data and Code Availability. All datasets and code are open-source and reproducible at `github.com/ShiroMorphy/dynamic-causal-emergence`.

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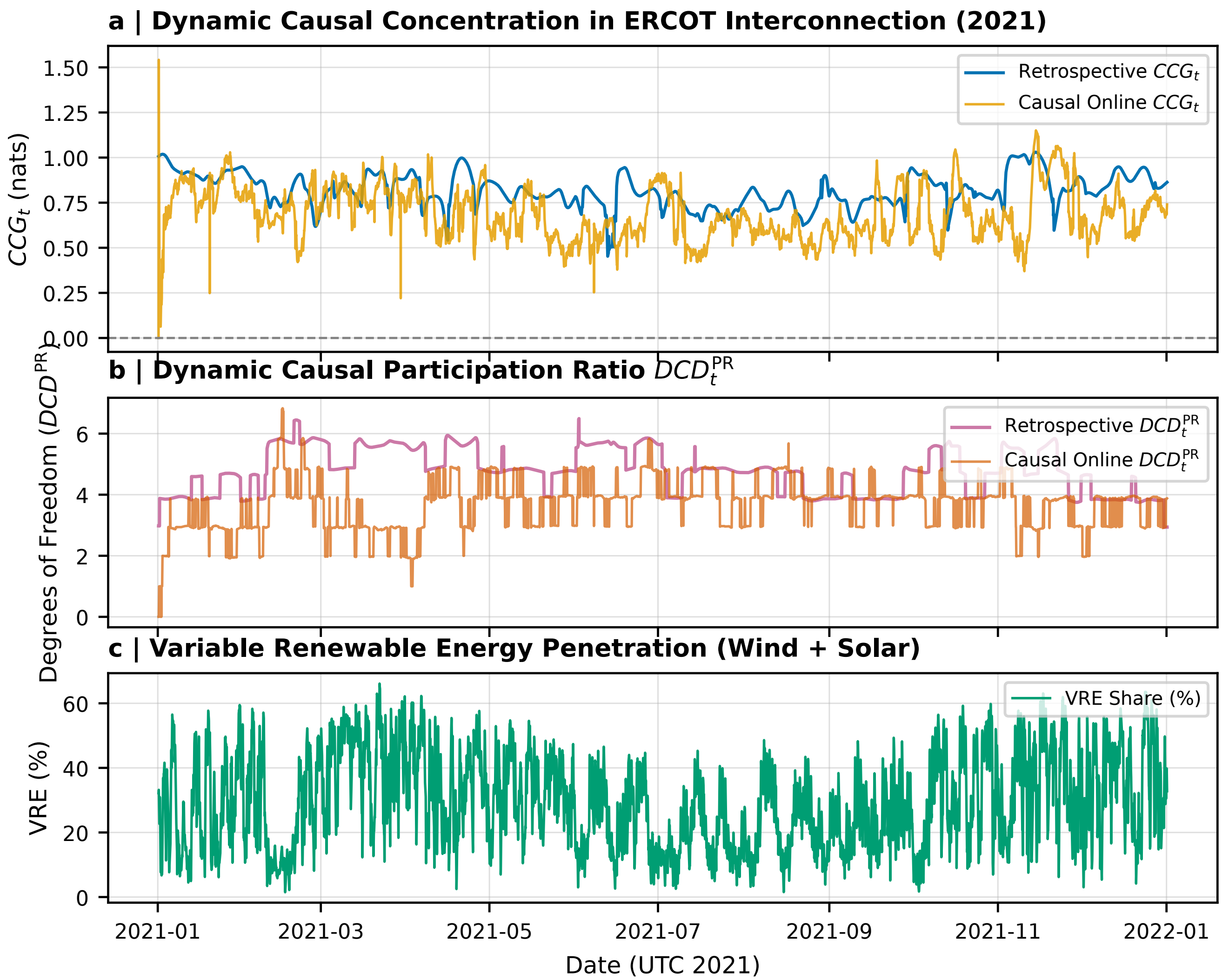

Figure 3: **Empirical Dynamic Causal Concentration and Dimensionality in ERCOT (EIA-930, 2021).** **a**, Hourly trajectory of Causal Concentration Gain CCG_t in ERCOT ($p = 9$, fuel-extended microstate), comparing retrospective symmetric filtering (blue) against causal one-sided online tracking (yellow). **b**, Dynamic Causal Participation Ratio DCD_t^{PR} tracking continuous effective degrees of freedom over time. **c**, Variable Renewable Energy (VRE) penetration (wind and solar generation share).

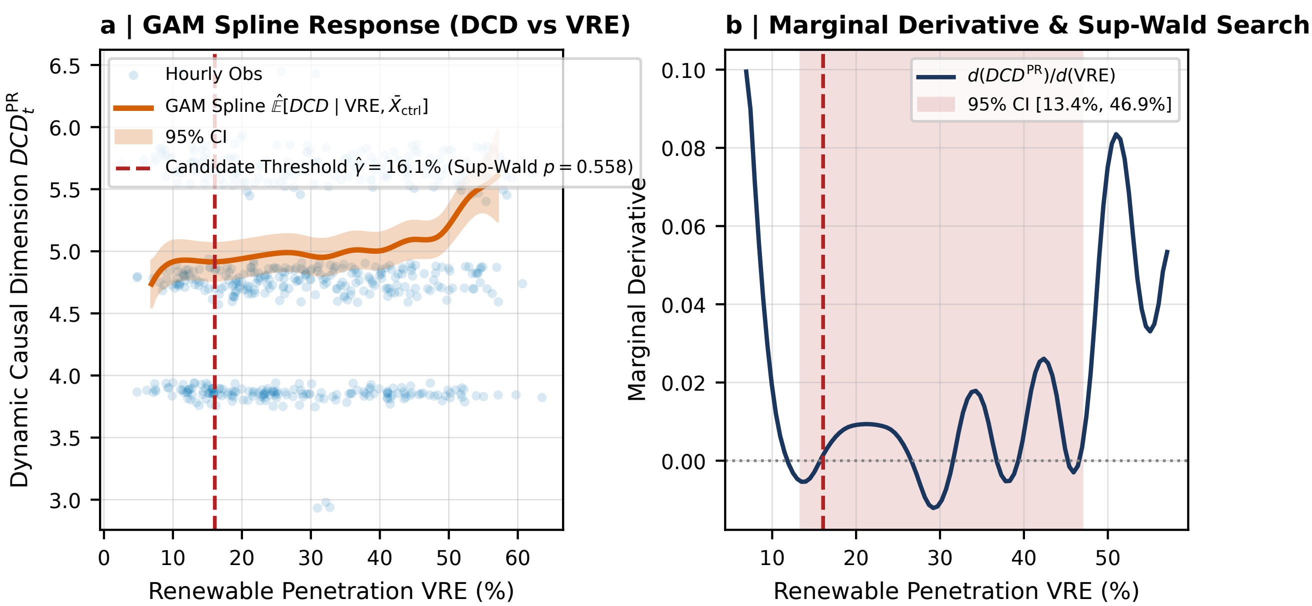

Figure 4: **Nonlinear Coordination Associated with Variable Renewable Generation.** **a**, Hourly observations and Generalized Additive Model (GAM) cyclic P-spline $\hat{\mathbb{E}}[DCD \mid \text{VRE}, \bar{X}_{\text{ctrl}}]$ with 95% bootstrap confidence band for ERCOT, showing smooth coordination along renewable penetration, with empirical threshold search candidate $\hat{\gamma} = 16.1\%$ (Hansen (1996) supremum-Wald block bootstrap test $p = 0.5582$, failing to reject the null hypothesis of smooth continuous adaptation). **b**, Marginal sensitivity derivative $d(DCD^{\text{PR}})/d(\text{VRE})$ with candidate threshold location 95% confidence set $[13.4\%, 46.9\%]$, conditional on the segmented regression specification.

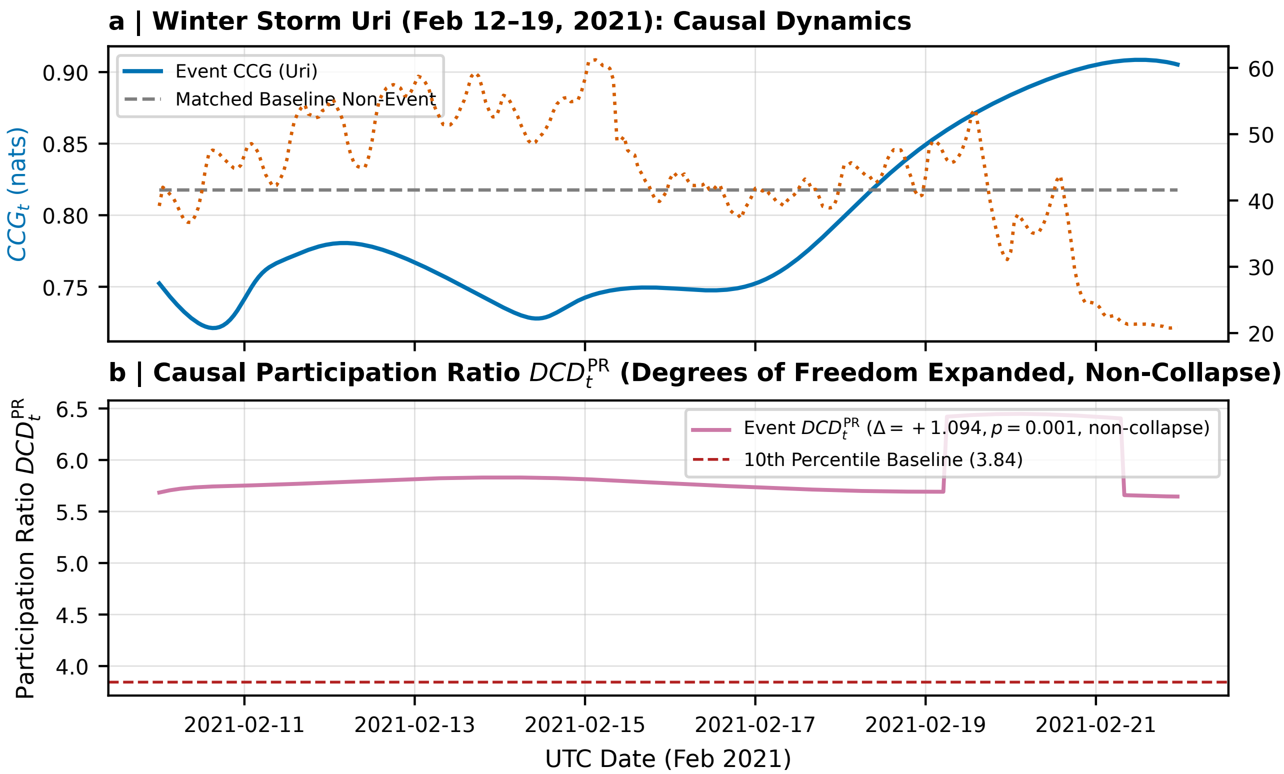

Figure 5: **Macroscopic Causal Reorganization During Winter Storm Uri (February 2021).** **a**, Hourly trajectory of Causal Concentration Gain CCG_t during Winter Storm Uri (solid blue) compared against matched non-crisis historical baseline (dashed grey; event mean 0.777 vs baseline 0.818, $\Delta CCG = -0.040, p_{\text{block}} = 0.5764$), overlaid with ERCOT Net Load (GW, right axis). **b**, Dynamic Causal Participation Ratio DCD_t^{PR} dynamics during the freeze, showing that effective causal degrees of freedom were maintained throughout the crisis ($\Delta DCD^{\text{PR}} = +1.094, p_{\text{block}} = 0.0010$, remaining strictly above the 10th percentile baseline threshold 3.84 with 0% collapse rate).

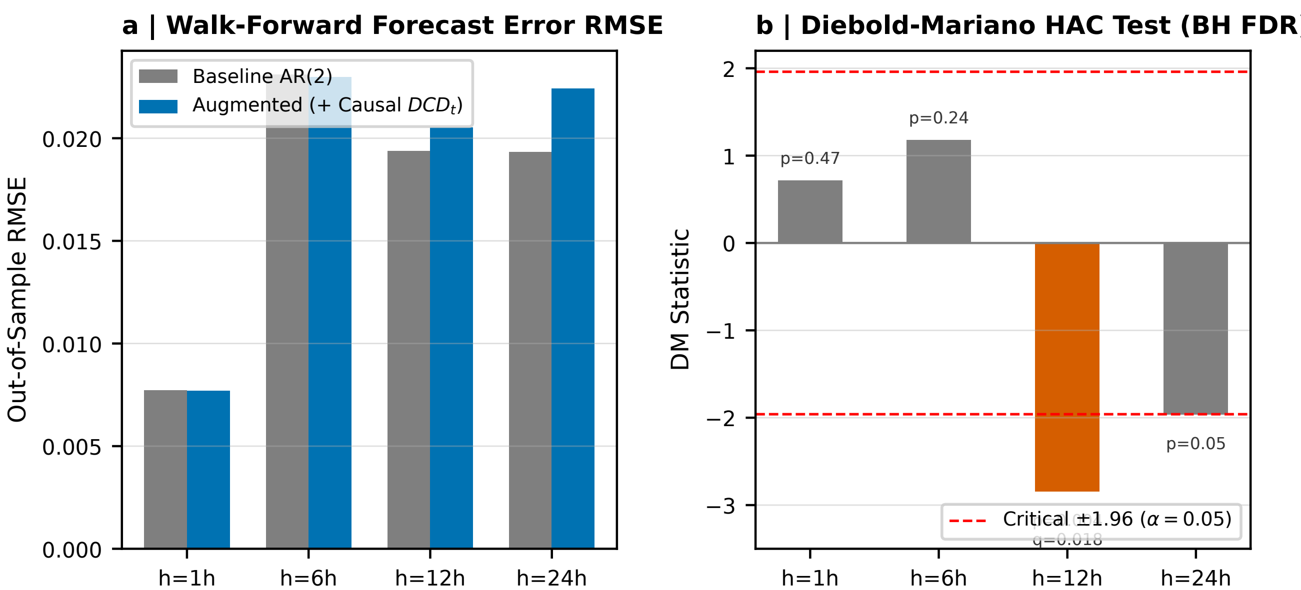

Figure 6: **Out-of-Sample Operational Predictability and Multiplicity-Corrected Diebold-Mariano Tests.** **a**, Lookahead-free walk-forward demand forecast error RMSE for baseline AR(2) vs augmented (+ Causal DCD_t) across forecasting horizons $h \in \{1, 6, 12, 24\}$ hours. **b**, Diebold-Mariano test statistics with Newey-West HAC covariance and Benjamini-Hochberg FDR multiplicity correction ($h = 1$ h: $p = 0.474, q = 0.474$; $h = 6$ h: $p = 0.238, q = 0.317$; $h = 12$ h: ΔRMSE $= -5.92\%$, DM stat -2.85, $p = 0.0044, q = 0.0178$; $h = 24$ h: $p = 0.0495, q = 0.0990$). Although point forecast accuracy deteriorates at 12 h under FDR ($q = 0.0178$), causal macroscopic metrics do not improve linear point forecasts, establishing DCD as an interpretable structural coordination diagnostic rather than a generic linear predictor.